# Resilient Communication Networks

## Introduction To Resilience



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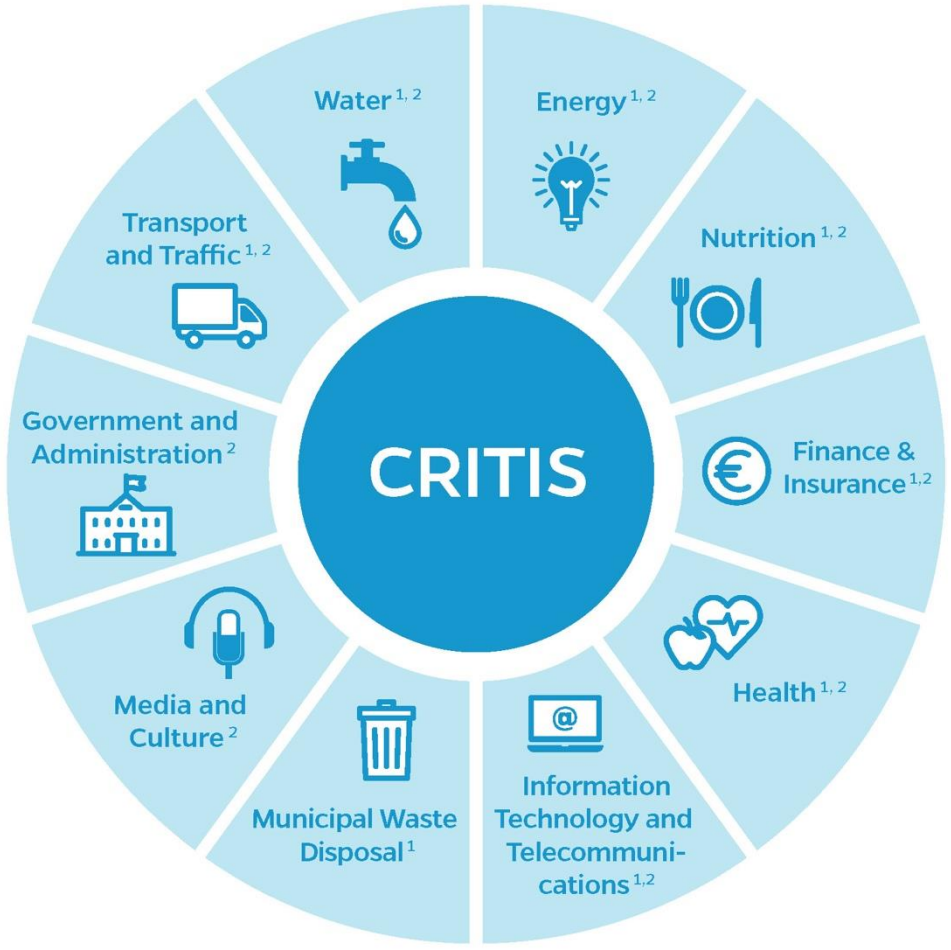
Disclaimer: some of these slides have been derived from Günter Schäfer's lecture „Protection of Communication Infrastructures", Mathias Fischers Lecture „Resilient Networks", Matthias Hollicks Lecture „ Ringvorlesung KRITIS", and Ralf Steinmetz' lecture "Communication Networks II"

# Why do we need resilience?

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## Critical infrastructures are increasingly reliant on Information and Communication Techology (ICT)

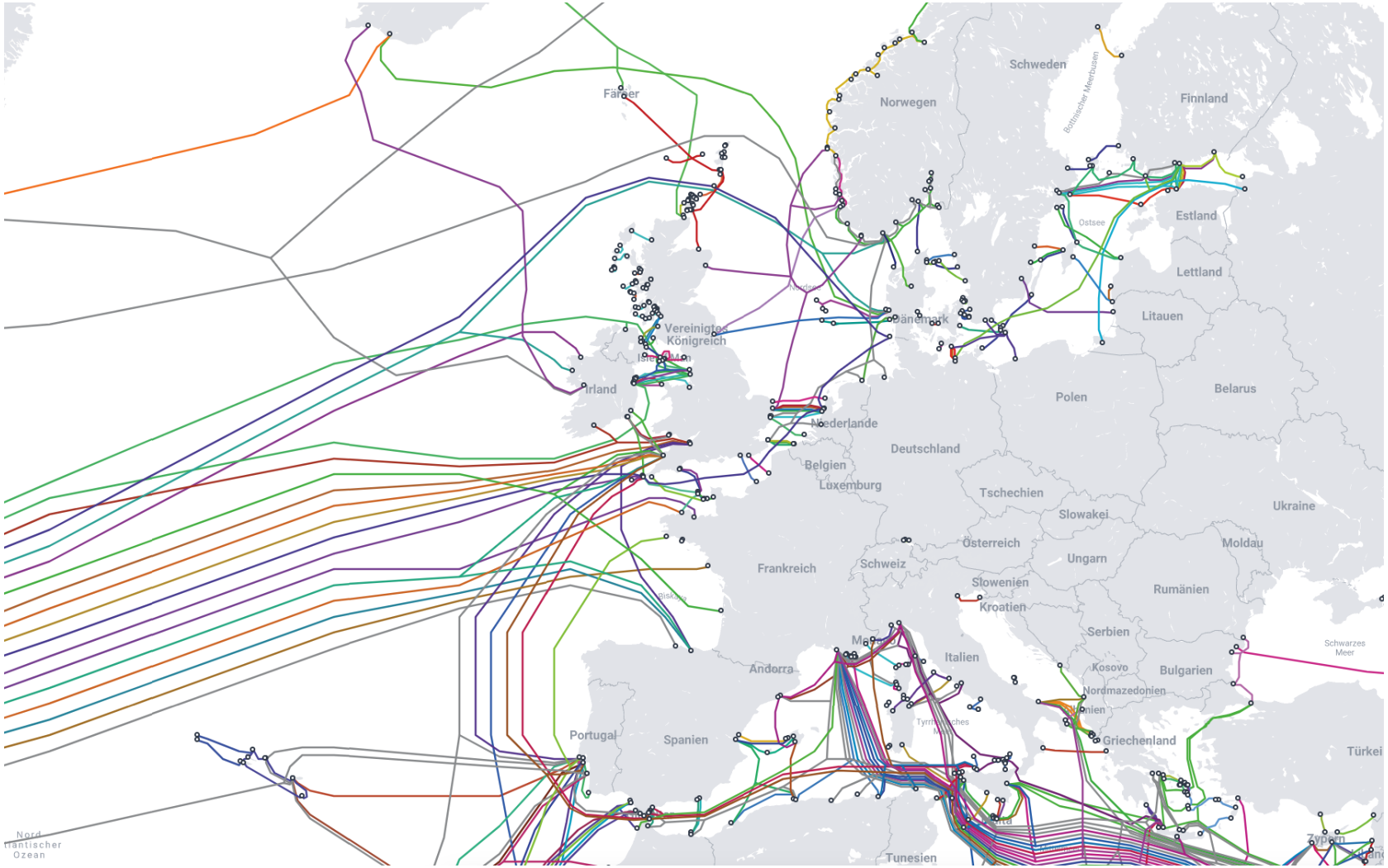
- The energy providers utilize ICT to coordinate the power grid
- The reachability of government organisations depends largely on ICT




[1] according to the BSIG "Federal Office for Information Security Act"
[2] according to the Federal-State Working Group

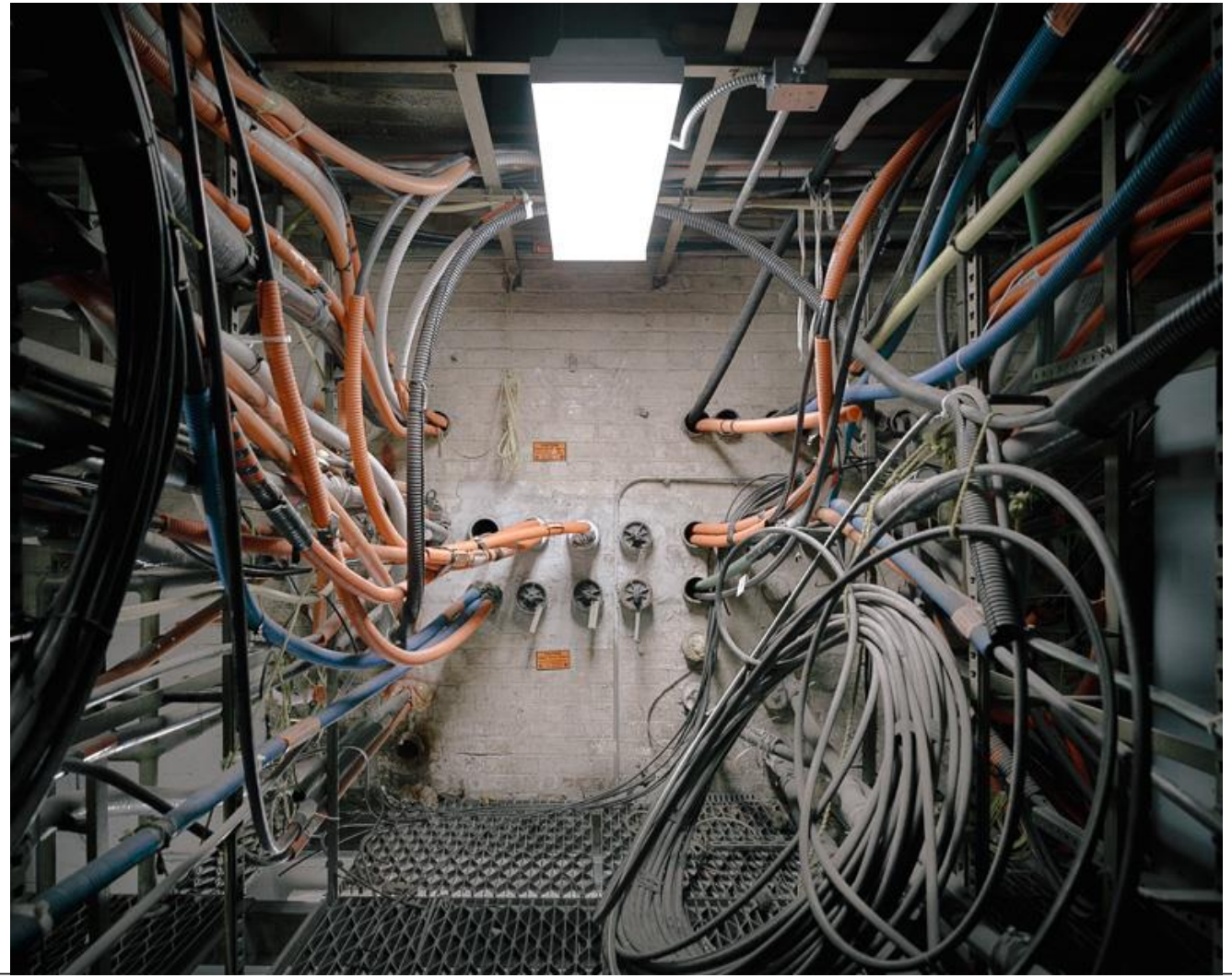
https://interflex.com/en/critical-infrastructures/

# Fiber Optics Connecting States




https://www.submarinecablemap.com/

# Somewhere in Manhattan…



http://www.petergarritano.com/the-internet.html

# Pinging columbia.edu

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```
traceroute: Warning: columbia.edu has multiple addresses; using 162.159.138.64
traceroute to columbia.edu (162.159.138.64), 64 hops max, 40 byte packets
 1  10.10.0.1 (10.10.0.1)  3.899 ms  5.136 ms  3.462 ms
 2  130.83.245.126 (130.83.245.126)  4.216 ms  3.795 ms  3.715 ms
 3  cores102x.net.hrz.tu-darmstadt.de (130.83.227.112)  3.987 ms  4.019 ms  3.954 ms
 4  spinehrz3-2.net.hrz.tu-darmstadt.de (130.83.227.51)  4.108 ms
    spinehrz4-2.net.hrz.tu-darmstadt.de (130.83.227.47)  3.967 ms  3.984 ms
 5  130.83.227.171 (130.83.227.171)  3.785 ms
    130.83.227.247 (130.83.227.247)  3.828 ms
    13
 6  13
 7  et                                                                    .5 ms
 8  et-1-2-1-400.core1.karo.tu.da.man-da.net (82.195.78.45)  4.299 ms  4.165 ms  4.081 ms
 9  et-1-0-0-0.core1.sm.tu.da.man-da.net (82.195.67.77)  4.191 ms  4.350 ms  4.144 ms
10  te-1-0-0-0.core1.fr5.eqx.f.man-da.net (82.195.80.63)  4.430 ms  4.821 ms
    te-0-0-6-0.core1.fr5.eqx.f.man-da.net (82.195.80.61)  8.667 ms
11  te-1-0-0-0.peer1.fr5.eqx.f.man-da.net (82.195.67.104)  4.937 ms  4.656 ms  4.947 ms
12  as13335.frankfurt.megaport.com (62.69.146.42)  5.590 ms  8.760 ms  6.031 ms
13  162.158.84.109 (162.158.84.109)  9.272 ms  9.774 ms
    162.158.84.55 (162.158.84.55)  18.506 ms
14  162.159.138.64 (162.159.138.64)  5.608 ms  4.792 ms  4.811 ms
```

Is that server located in the US?

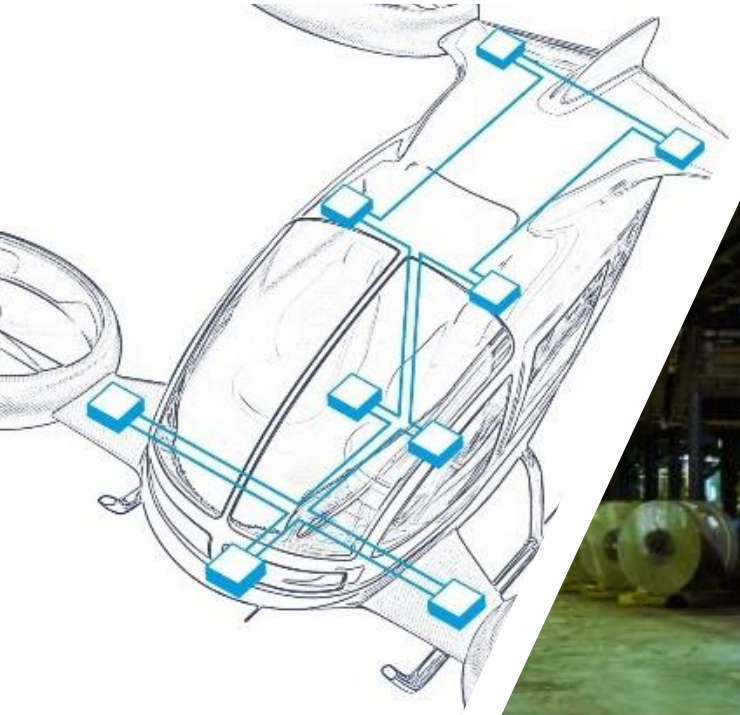














# … and there are other networks

# History of resilience
## Telegraph Networks

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### Earlier telegraphy networks were linear and centralized

- Point-to-point connections
- Communication via morse telegraphs

### Failures were often catastrophic

- A simple cable cut at any point was rendering the communication useless
- The centralized architecture offered simple ways of disrupting communication

### Redundancy was (in large parts) not existent

- Backup lines were often only introduced during conflicts

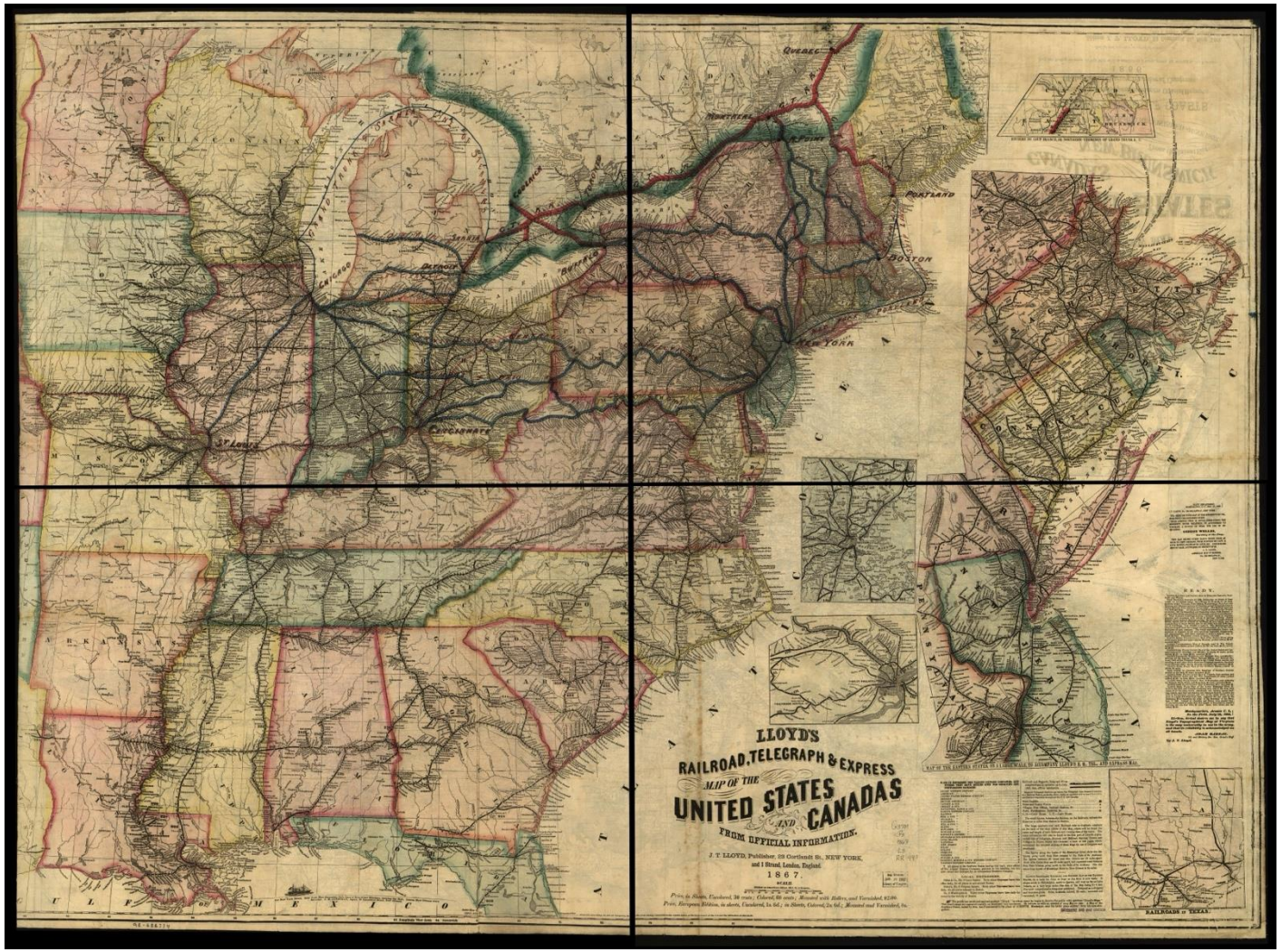



Image source: Wikimedia Commons, Library of Congress

# History of resilience
## Telephone Networks

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### First telephones were also sold pairwise

- Direct connection between households
- No flexibility



### Later, Telephone switches reduced complexity of phone network

- Line from each phone to central switchboard
- Long distance lines between switchboards

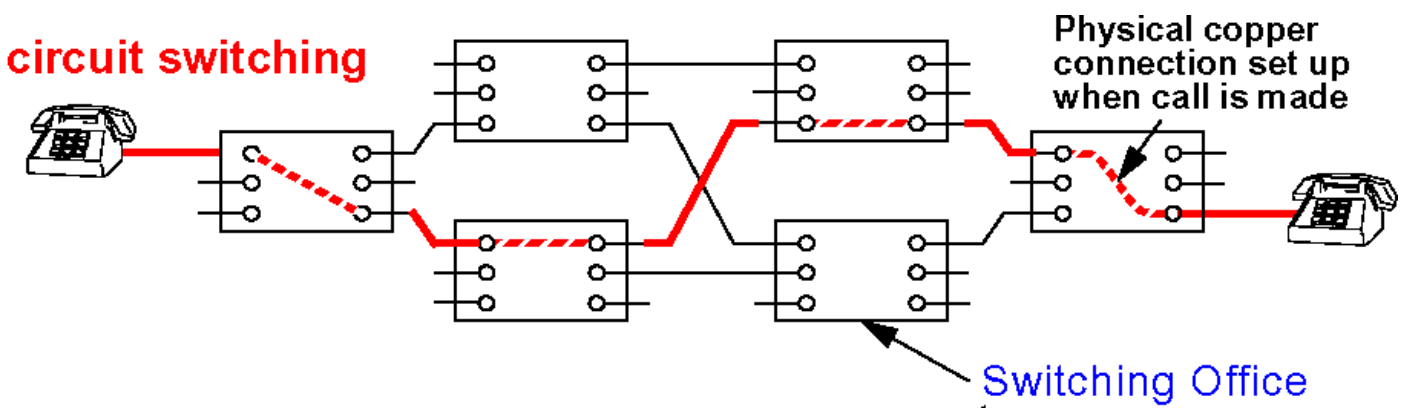






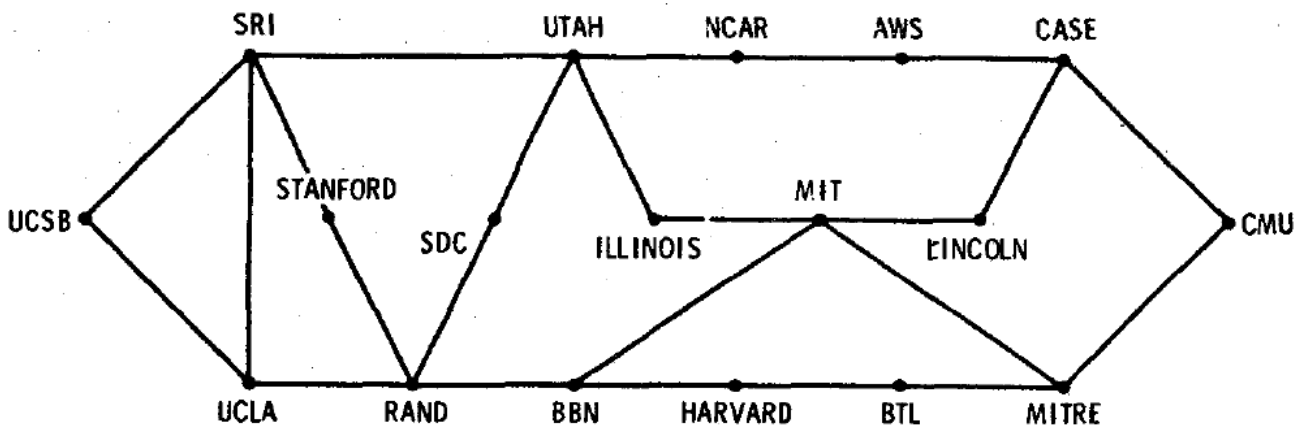
Image source: Wikimedia Commons

# History of resilience
## The Internet

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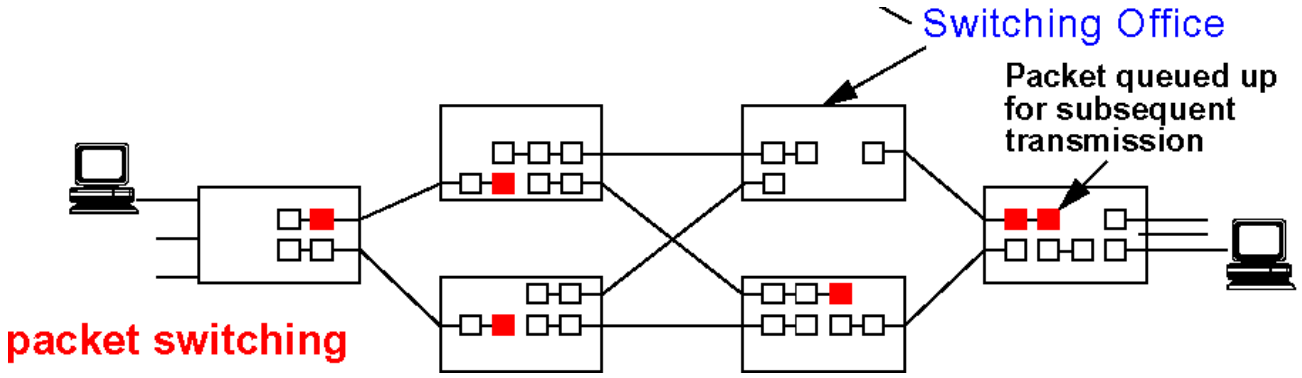
### ARPANET as the predecessor of the Internet

- Build specifically for resilience during the cold war
- Idea of packet switching to avoid centralized vulnerabilities
- Demonstrated survivability in case of node/link failure




### TCP/IP later introduced to provide reliable end-to-end communication

- Error checking, retransmissions, flow control



# In the last and next Years

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**Mobile telephony**

- (4G) 5G .. 6G

**Web (last century)**

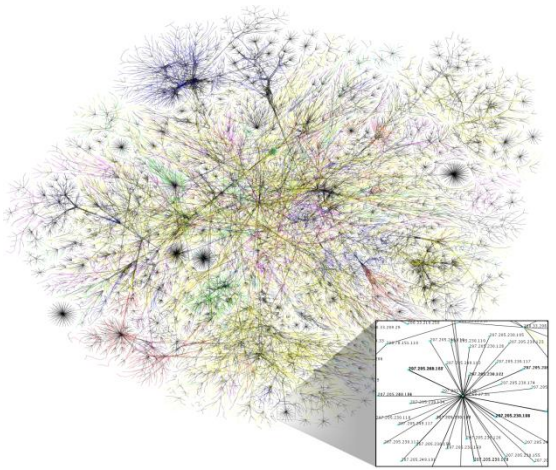
**Peer-to-Peer**

**Cloud Services**

- like AWS, Google Cloud…

**and**

- Web services
- Twitter, WhatsApp
- Online social networks
- …





# In the next Years: Our real networked World

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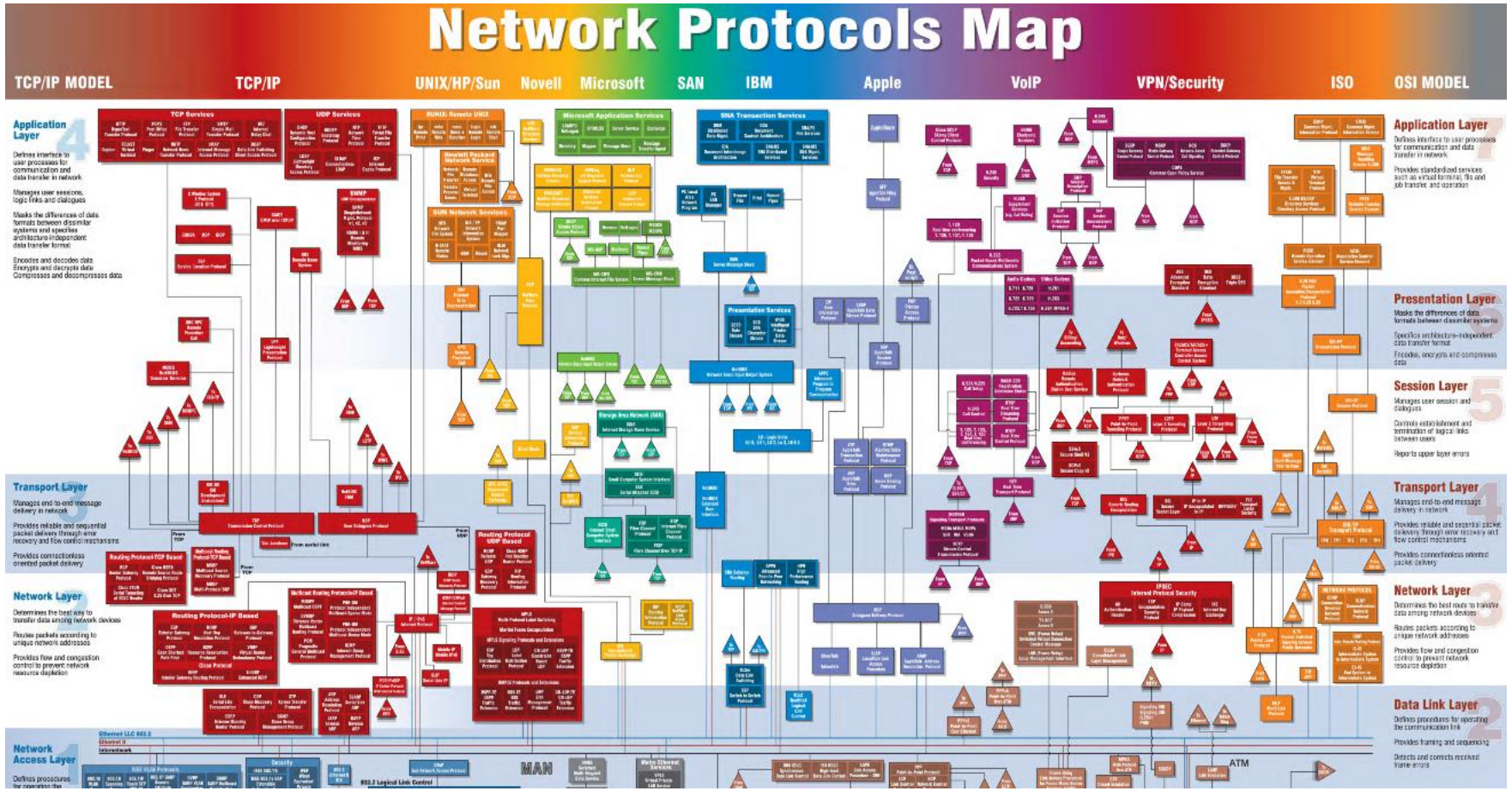
**Which changes have happened due to the Internet changed over the last 20 years?**

**Where does the Internet influence our life today?**

**How will the Internet evolve?**
**What will be different due to the future Internet?**

# Mechanisms, Protocols, Services, Architecture

not relevant for the exam

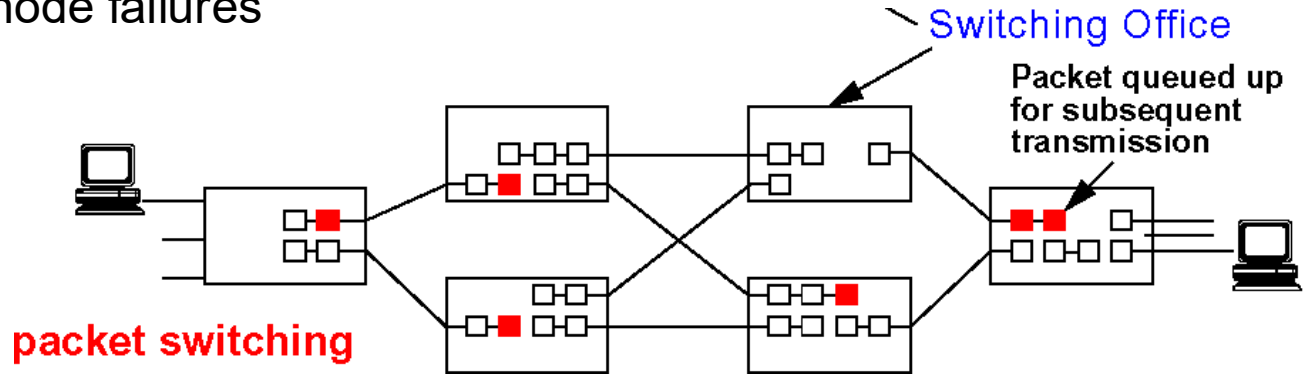


# History of resilience
## Internet Protocols

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**Protocols and mechanisms used on the Internet often exhibit a high degree of resilience**

- Border-Gateway protocol (BGP) finds routes on the Internet without central coordination
- Domain-Name System (DNS) is inherently distributed
- Packet switching allows for recovery of link and node failures




Are we done?

# Facebook Incident

**At 4th of October 2021, a command issues as part of a routine maintenance accidentally disconnected all Facebook datacenters**

- As a result, Facebooks did not advertise BGP routes to their datacenters anymore
- For a few hours, Facebook literally disappeared from the Internet, erasing every trace of itself



**Communication within the company was affected as well**

- Employees could not communicate with each other over Facebooks internal network
- Some badge-protected doors prevented access to Facebook buildings

**After employees gained physical access to Facebook facilities, things returned to normal**

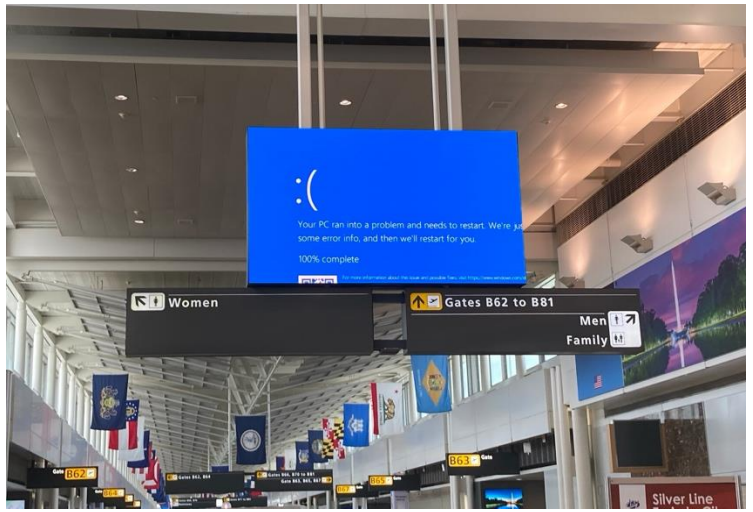
https://engineering.fb.com/2021/10/05/networking-traffic/outage-details/

# Crowdstrike Incident

**At 19th of July 2024, a faulty update of Crowdstrikes Falcon Sensor led to one of the largest outages in IT history**

- The update of the detection rule `C-00000291*.sys` led to the bluescreen of death
- 8.5 Million Windows-based devices were rendered unusable
- Airports, hospitals, supermarkets, and banks were affected

**Manual intervention was necessary to fix the issue, costing thousands of man hours**






# Fiber Incident of Deutsche Bahn

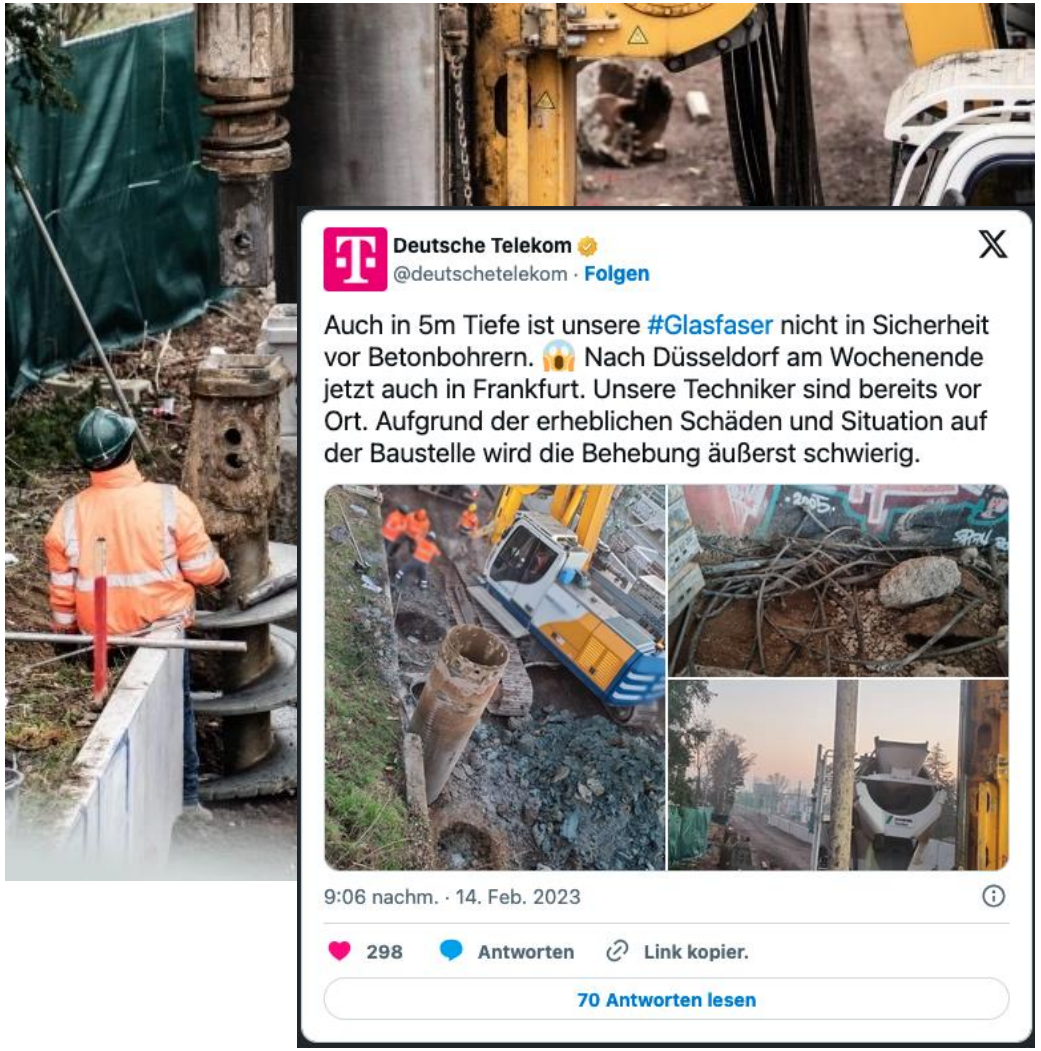
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**In 2023, Deutsche Bahn accidentally destroyed a fiber from Deutsche Telekom**

- Disconnect of Frankfurt Airport
- Additional disruptions of communication in Frankfurt city

**However: Communication was designed to be redundantly available**

- Routing was not efficiently managed via the backup line, causing additional disruptions
- Backup line failed a few hours after the main system



# Shocks Affecting ICT

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**Traditional or risk-based management insufficient**



Natural disaster



Cyberevents



Blackout



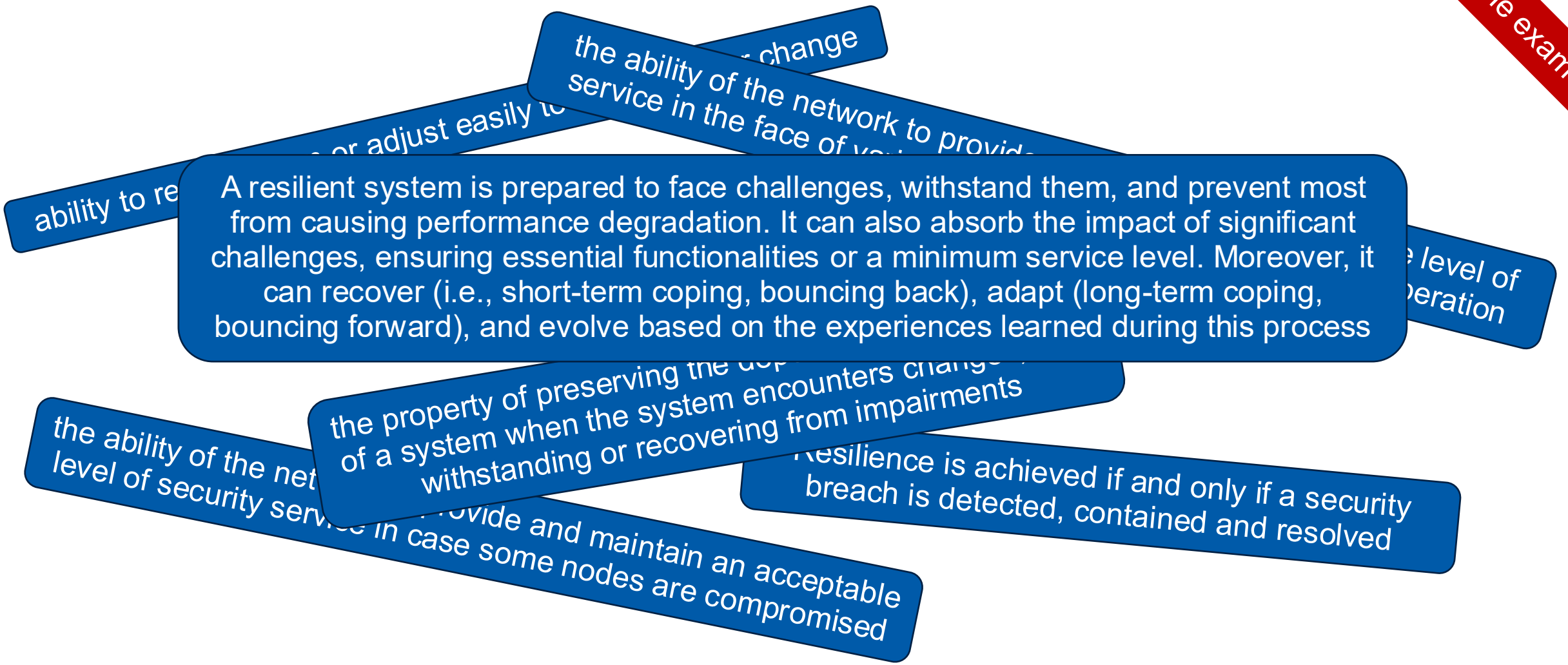
Terror



”Allgefahren“

**Increasing complexity of (unexpected) crises in today‘s complex and networked societies**

# Resilience Definitions




not relevant for the exam

Khaloopour et al., "Resilience-by-Design in 6G Networks: Literature Review and Novel Enabling Concepts," in IEEE Access, vol. 12, pp. 155666-155695, 2024, doi: 10.1109/ACCESS.2024.3480275.

# Resilience in Communication Networks



- **Absorption through resistance, elasticity and adaptability**
- **Recovery by returning to normal state**
- **Transformation through the transfer of functionality**
- **Sustainability through learning from experience and maturation / further development**

**Resilience describes the ability of a system**

**(1) to absorb crises,**
**(2) to recover from them quickly and sustainably,**
**(3) to achieve comparable or new functionality through transformation**

# Now That You Know What Resilience IS, …

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… how can we **increase the resilience** of a communication network?

… when is a communication network **not resilient**?

… how can we **measure** resilience?

… what **incidents/ threats** do we have to account for? Can a system even be resilient to everything?

… what are the **downsides** of a highly resilient system?

… is there a **master plan** of how a resilient system **reacts**?

… is resilience a **static** feature or a **process**?

…

# ResiliNets Initiative

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Tries to **systematically analyse resilience** in communication networks

## Main works

- Sterbenz et al. (2010). Resilience and Survivability in Communication Networks: Strategies, Principles, and Survey of Disciplines. Computer Networks. 54(8). pp. 1245-1265. doi: 10.1016/j.comnet.2010.03.005.
- A. Jabbar (2010). A Framework to Quantify Network Resilience and Survivability. Dissertation. Kansas School of Engineering, University of Kansas.
- ResiliNets Wiki: https://resilinets.org/

# ResiliNets Initiative

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## Contributions

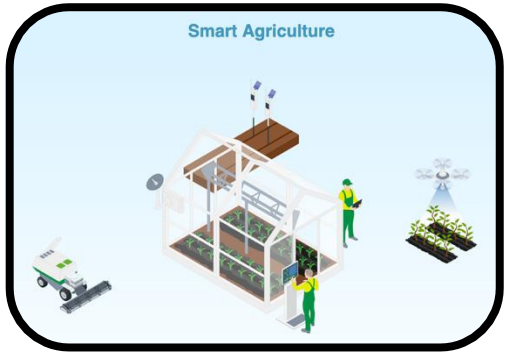
- **Resilience disciplines** that
  - assess requirements of communication networks that can tolerate challenges
  - define measurable properties of resilient communication networks
- **Challenge classification**
- **Resilience strategy** that
  - helps to design communication networks that rapidly adapt to challenges and attacks and at the same time maintain an acceptable level of service
  - enables longer-term evolution of communication networks
- **Design principles** including prerequisites, tradeoffs, enablers, and behavior
- Analysis concepts to **measure and quantify network resilience**

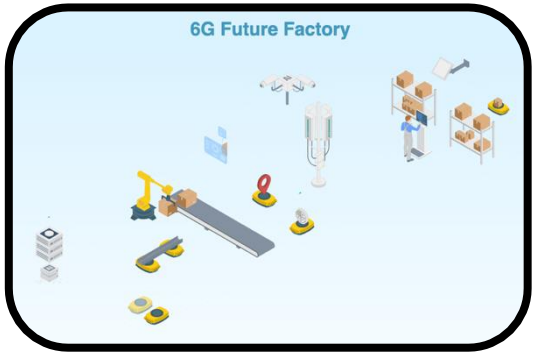
# BMBF 6G-Plattform

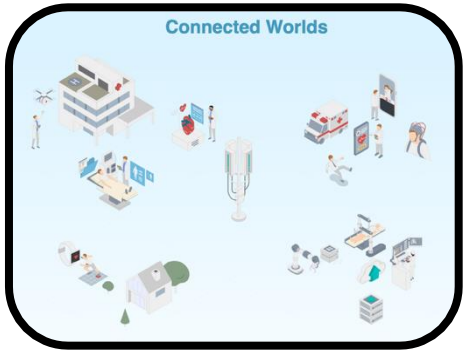
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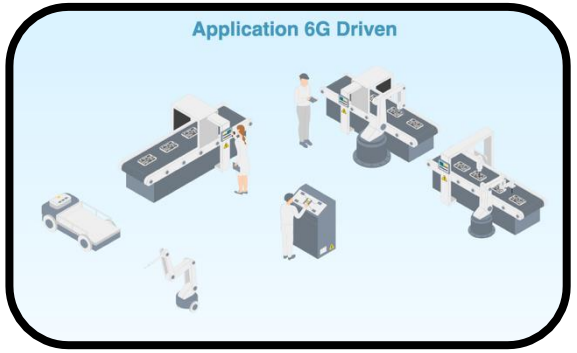
## Resilience is an important enabler for future 6G networks

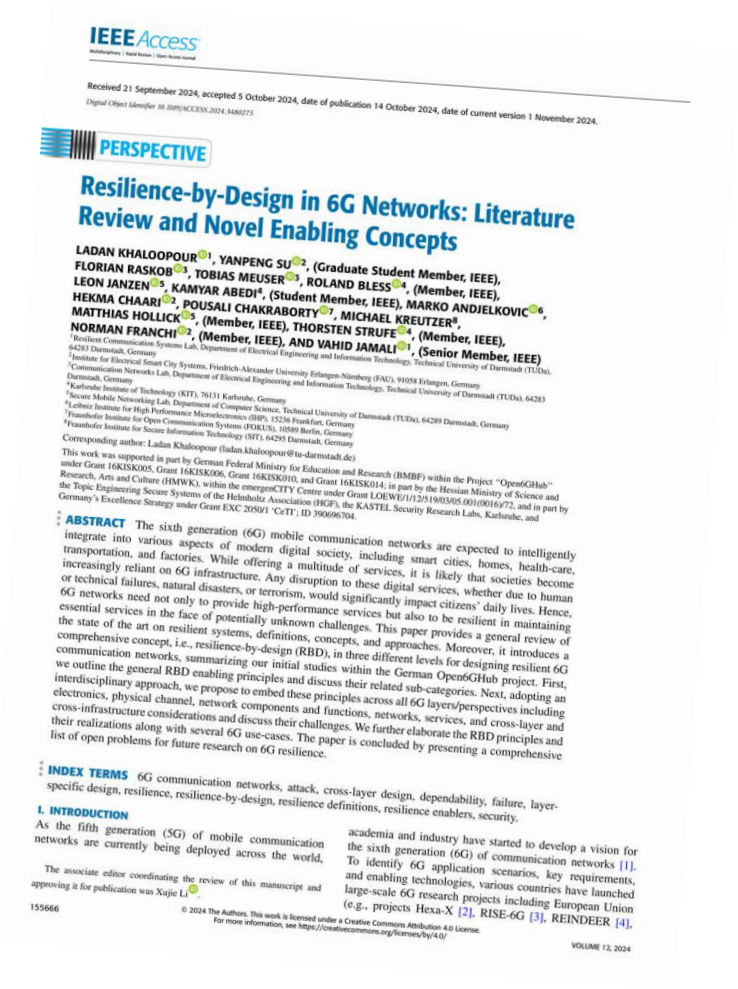
- Increasing utilization of cellular networks in society
- Increasing utilization of 5G and beyond for factory automation



IEEE Access

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PERSPECTIVE

**Resilience-by-Design in 6G Networks: Literature Review and Novel Enabling Concepts**

LADAN KHALOOPOUR[1], YANPENG SU[2], (Graduate Student Member, IEEE), FLORIAN RASKOB[3], TOBIAS MEUSER[3], ROLAND BLESS[4], (Member, IEEE), LEON JANZEN[5], KAMYAR ABEDI[4], (Student Member, IEEE), HEKMA CHAARI[2], POUSALI CHAKRABORTY[7], MARKO ANDJELKOVIC[6], MATTHIAS HOLLICK[5], (Member, IEEE), THORSTEN STRUFE[4], (Member, IEEE), NORMAN FRANCHI[2], (Member, IEEE), AND VAHID JAMALI[1], (Senior Member, IEEE)

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**ABSTRACT** The sixth generation (6G) mobile communication networks are expected to intelligently integrate into various aspects of modern digital society, including smart cities, homes, health-care, transportation, and factories. While offering a multitude of services, it is likely that societies become increasingly reliant on 6G infrastructure. Any disruption to these digital services, whether due to human or technical failures, natural disasters, or terrorism, would significantly impact citizens' daily lives. Hence, 6G networks need not only to provide high-performance services but also to be resilient in maintaining essential services in the face of potentially unknown challenges. This paper provides a general review of the state of the art on resilient systems, definitions, concepts, and approaches. Moreover, it introduces a comprehensive concept, i.e., resilience-by-design (RBD), in three different levels for designing resilient 6G communication networks, summarizing our initial studies within the German Open6GHub project. First, we outline the general RBD enabling principles and discuss their related sub-categories. Next, adopting an interdisciplinary approach, we propose to embed these principles across all 6G layers/perspectives including electronics, physical channel, network components and functions, networks, services, and cross-layer and cross-infrastructure considerations and discuss their challenges. We further elaborate the RBD principles and their realizations along with several 6G use-cases. The paper is concluded by presenting a comprehensive list of open problems for future research on 6G resilience.

**INDEX TERMS** 6G communication networks, attack, cross-layer design, dependability, failure, layer-specific design, resilience, resilience-by-design, resilience definitions, resilience enablers, security.

**I. INTRODUCTION**

As the fifth generation (5G) of mobile communication networks are currently being deployed across the world, academia and industry have started to develop a vision for the sixth generation (6G) of communication networks [1]. To identify 6G application scenarios, key requirements, and enabling technologies, various countries have launched large-scale 6G research projects including European Union (e.g., projects Hexa-X [2], RISE-6G [3], REINDEER [4],

The associate editor coordinating the review of this manuscript and approving it for publication was Xujie Li.

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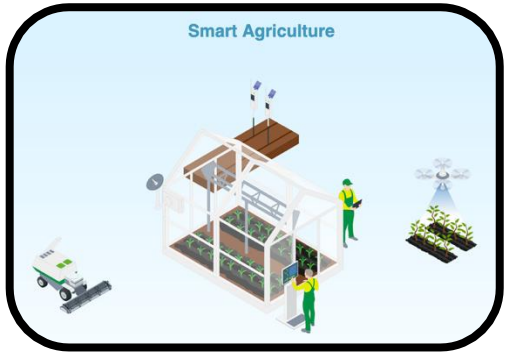
Taken from https://openlabs.open6ghub.de/#/worlds

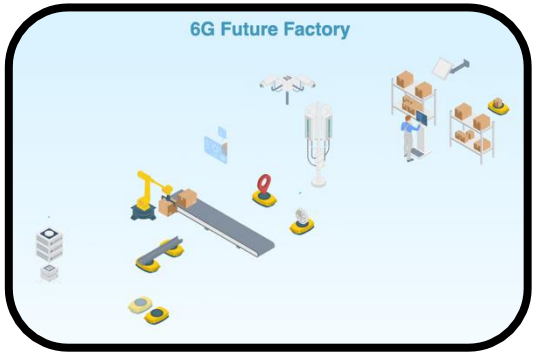
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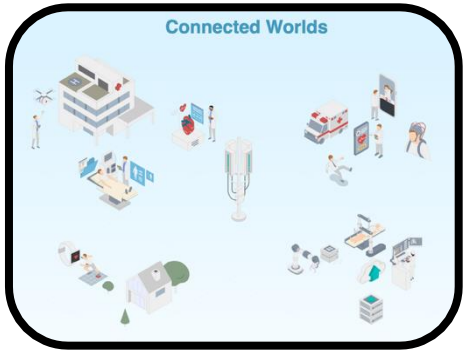
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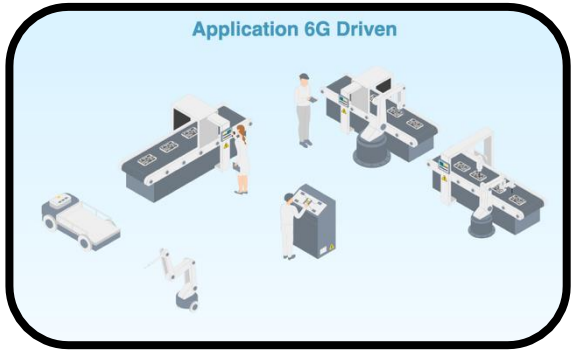
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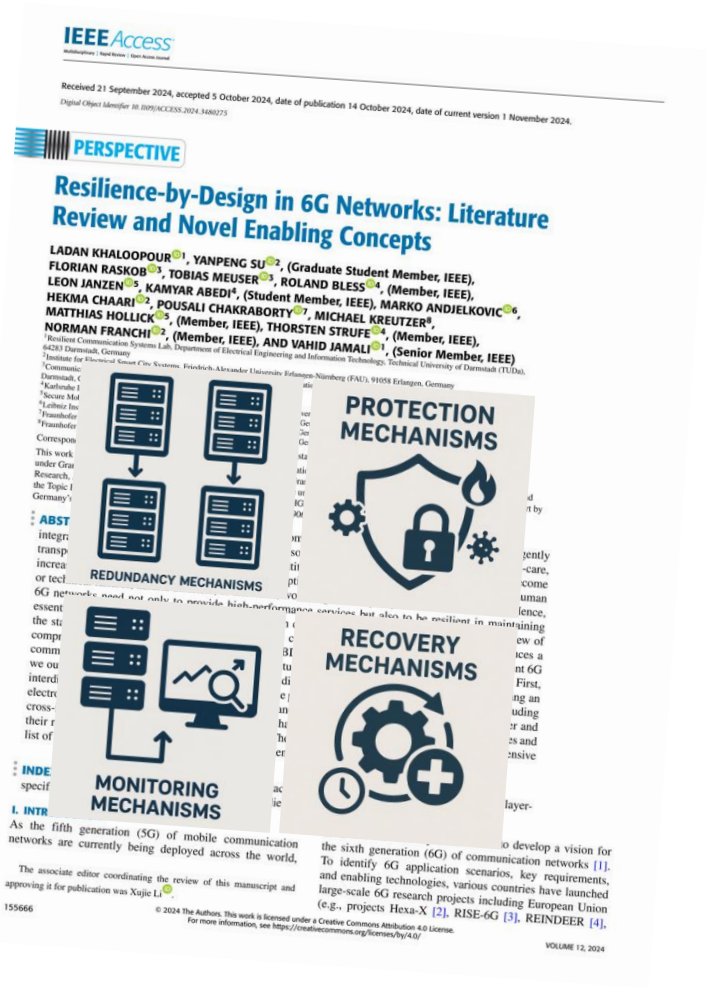
- Increasing utilization of cellular networks in society
- Increasing utilization of 5G and beyond for factory automation



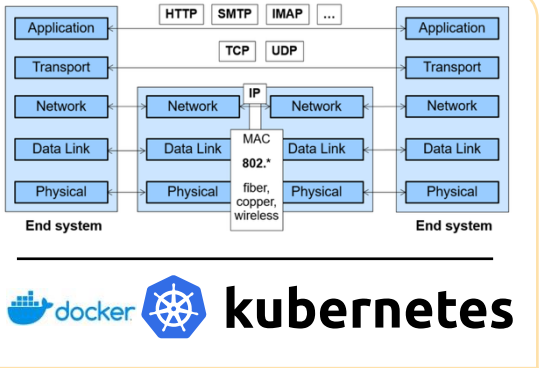



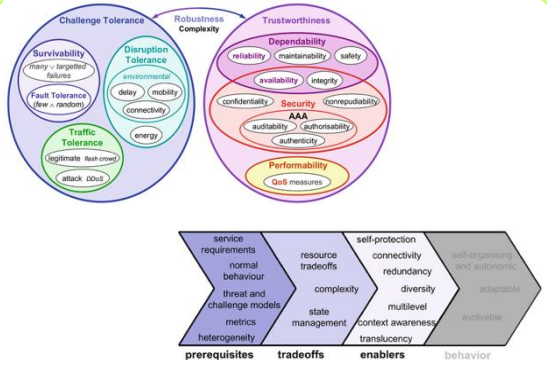




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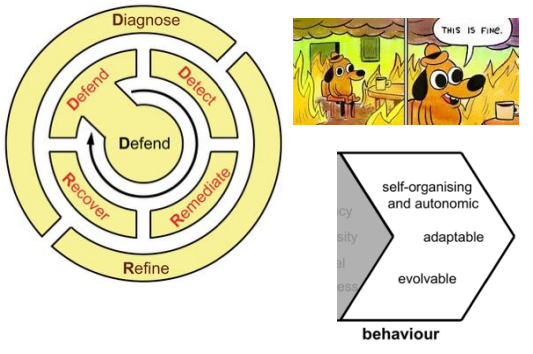
# Lecture Content Overview



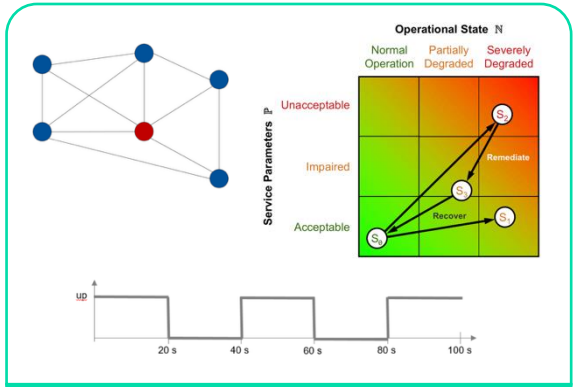

Introduction To Communication Networks and Kubernetes




Designing Resilient Communication Networks




Resilient Behavior




Resilience Metrics




Guest Lectures: Current Topics on Resilience

# Kubernetes Labs

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- Learn basic containerization and Kubernetes concepts
- Apply lecture content to practical problems
- Learn about 3-tier applications, RESTful APIs, full-stack implementation, web development, …
- Play your very own self-hosted flappy bird game and have fun ☺



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